

Milestone 4 Technical Summary

Architecture and End-to-End Behavior

The system now implements two real workflows. Search mode retrieves and ranks opportunities without pretending the search sentence is a posting. Posting mode classifies a job description, retrieves comparable evidence, and applies deterministic missing-data and authorization gates.

Layer	Method	Business control
Mode	search vs. posting	Prevents task confusion
Retrieval	80k sparse index + boosts + dedupe	Inspectable candidates
Decision	word+character TF-IDF LinearSVC	0.984 macro F1
Policy	missing fields + authorization evidence	Unclear / Hold
Response	evidence-bound template	No unsupported facts

Evaluation vs. Baseline

System	Rows	Accuracy	Macro F1
M2 same-subset baseline	4,000	-	0.791
M3 true PEFT LoRA	4,000	0.875	0.873
M4 trained decision layer	4,000	0.984	0.984

The same-subset macro-F1 gain is +0.192. The model was trained on 77,135 rows after exact-text deduplication, not only the 2,000 rows used by M3 LoRA.

Robustness Audit

The original random split contains 1,328 exact validation texts that also appear in training. After controlling for this, macro F1 remains 0.979 on 18,672 novel-text rows and 0.980 on 5,570 rows from unseen companies.

Grounded Workflow Evidence

Case	System result	Operational response
entry level analytics search	N/A (search)	Advisor review before forwarding
senior engineering posting	low_fit	Advisor review before forwarding
authorization restriction posting	high_fit	Hold - explicit authorization restriction requires advisor verification
thin metadata posting	unclear	Advisor review before forwarding

Risk and Grounding Analysis

Risk 1 is authorization misinformation. Role fit and authorization are separate: explicit negative language forces a Hold, and missing evidence never becomes eligibility. Risk 2 is weak-label overconfidence. The high score measures agreement with transparent rules, so the system preserves human review and reports novel-text and unseen-company slices rather than relying on a random split alone.

Data and Model Recommendation

Do not expand the public weak-label corpus beyond 100,000 rows yet, and do not train a larger transformer simply for a higher weak-label score. The learning curve shows that the available training split is sufficient. The next phase should collect 500-1,000 advisor-reviewed labels and several hundred postings with real authorization language, then compare the linear model and LoRA on that human test set.

Deployment Boundary

Use the prototype for first-pass prioritization. Search results, low-margin classifications, missing fields, and any authorization-sensitive case remain advisor-confirmed. No automated forwarding, rejection, or immigration decision is permitted.